

# Construction Management

➤ Monday, Wednesday and Friday (6 periods per week)

## **Book suggested:**

- ❖ Construction Project Management: By K.K Chitkara
- ❖ Construction Management and Machinery : BL Gupta, Gupta, Amit
- ❖ Construction planning Equipment and Methods: By R.L Peurifoy
- ❖ Construction Plant Excavating and Handling Equipment and Methods:  
By Harris, Frank
- ❖ Construction Management: By R.P. Adhikari
- ❖ Estimating, Valuation and Specifications: By G. S.Birdie
- ❖ Text Book of Construction Management: By S.K. Shrestha, R. K.  
Shrestha & S.K. Bhattarai



# Chapter 1: Construction Management Framework

## 1.1 Construction Landmarks:

- An object or feature of a landscape or town that is **easily seen and recognized from a distance**,
- Especially one that enables someone **to establish their location**
- World's famous Construction which **inspire us** are Construction Landmarks
- Land marks give us sense of place
- An event or discovery marking **an important stage or turning point in something** is land mark

Examples:

- Angkor: Siem Reap, Cambodia
- Sydney Opera House: Sydney
- The Eiffel Tower: Paris
- Taj Mahal: Agra, India
- Burj Khalifa: Dubai, United Arab Emirates
- Machu Picchu: Machu Picchu Archaeological Park, Peru
- The Great Wall of China: China

# Examples of Construction Land Marks

- World's tallest building
- World's longest bridge
- World's longest water tunnel
- World's longest railway land tunnel
- World's highest dam
- World's largest floor area of Building etc.

## 1.2 Scope of Construction Management

- Basic needs of human being: Food, Clothing and Shelter
- First Natural shelter is **Cave**
- Second is **plat form on trees** to protect from animals
- Third **small houses**
- Construction is the **indicator of civilization**
- Indicator of development of the nation is development of Infrastructure
- National development is directly linked with the development of construction activities
- A single individual can't do all the works

# Management and Construction Management

- **Management** is an organization in which **different class of people work together** to produce **qualitative** and **economical** products with the available **resources**
- **Management** includes the activities of setting the strategy of an organization and coordinating the efforts of its employees (or of volunteers) to accomplish its objectives through the application of available resources, such as financial, natural, technological, and human resources
- **Management** is the administration of an organization, whether it is a **business, a not-for-profit organization, or government body.**
- Huge amount of Resources (5Ms: Man, Money, Materials, Machine and Minute or Methodology) are used in construction industry
- Construction Management provides **modern tools and techniques** to **optimize the resources so, it has wide scope and importance.**

# Scope of Construction Management

- The main purpose of CM is to **control project time, cost, and quality**
- Reduce overall project costs
- contract negotiation
- Reducing change orders (work that is added to or deleted from the original scope of work of a contract, however, depending on the magnitude of the change, it may or may not alter the original contract amount and/or completion date.
- A change order may force a new project to handle significant changes to the current project
- Orders reduction in time
- Elimination of unnecessary scope and cost

# Construction Market

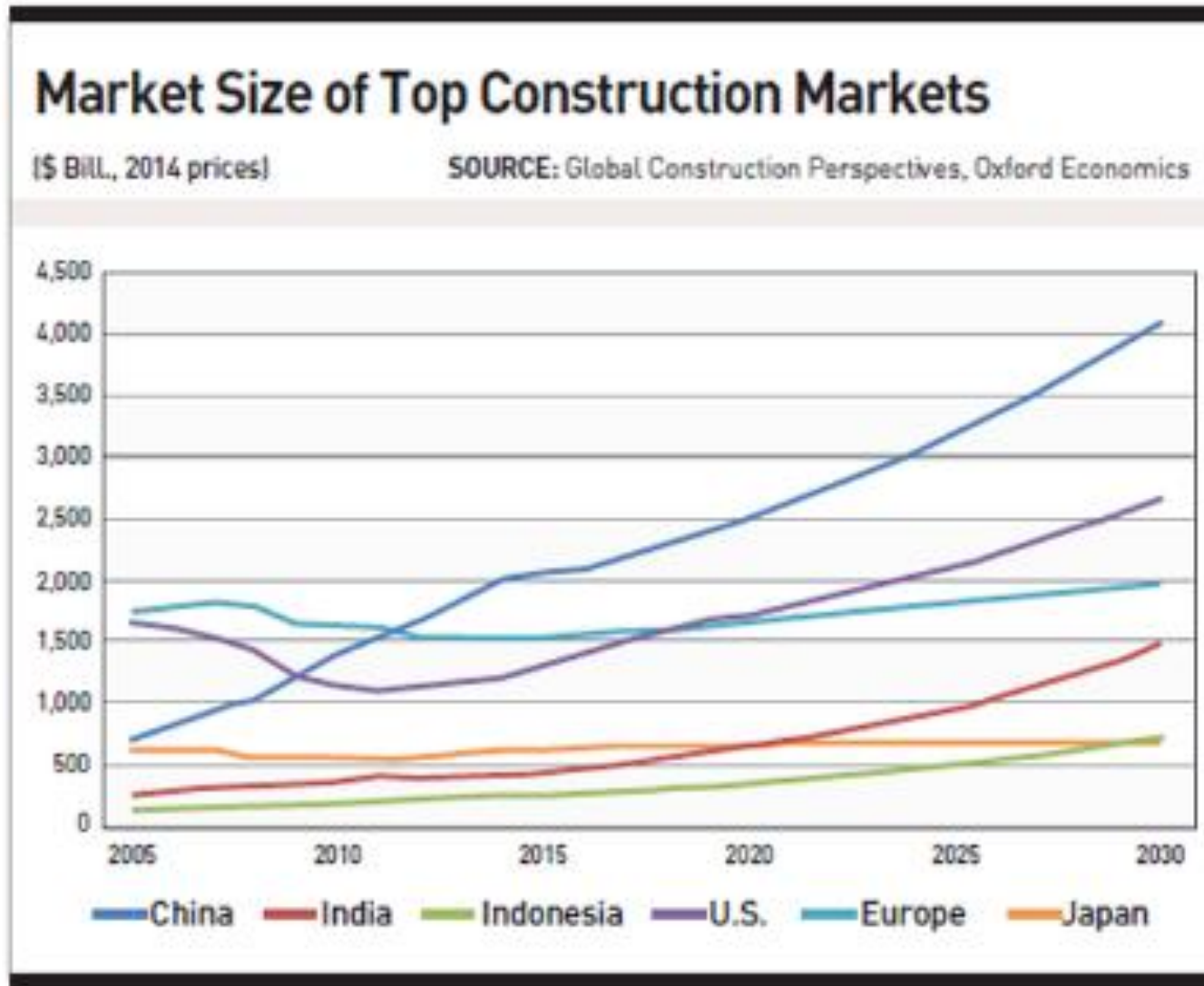
## **Construction market:**

- International
- Domestic market

## **Construction Market Size:**

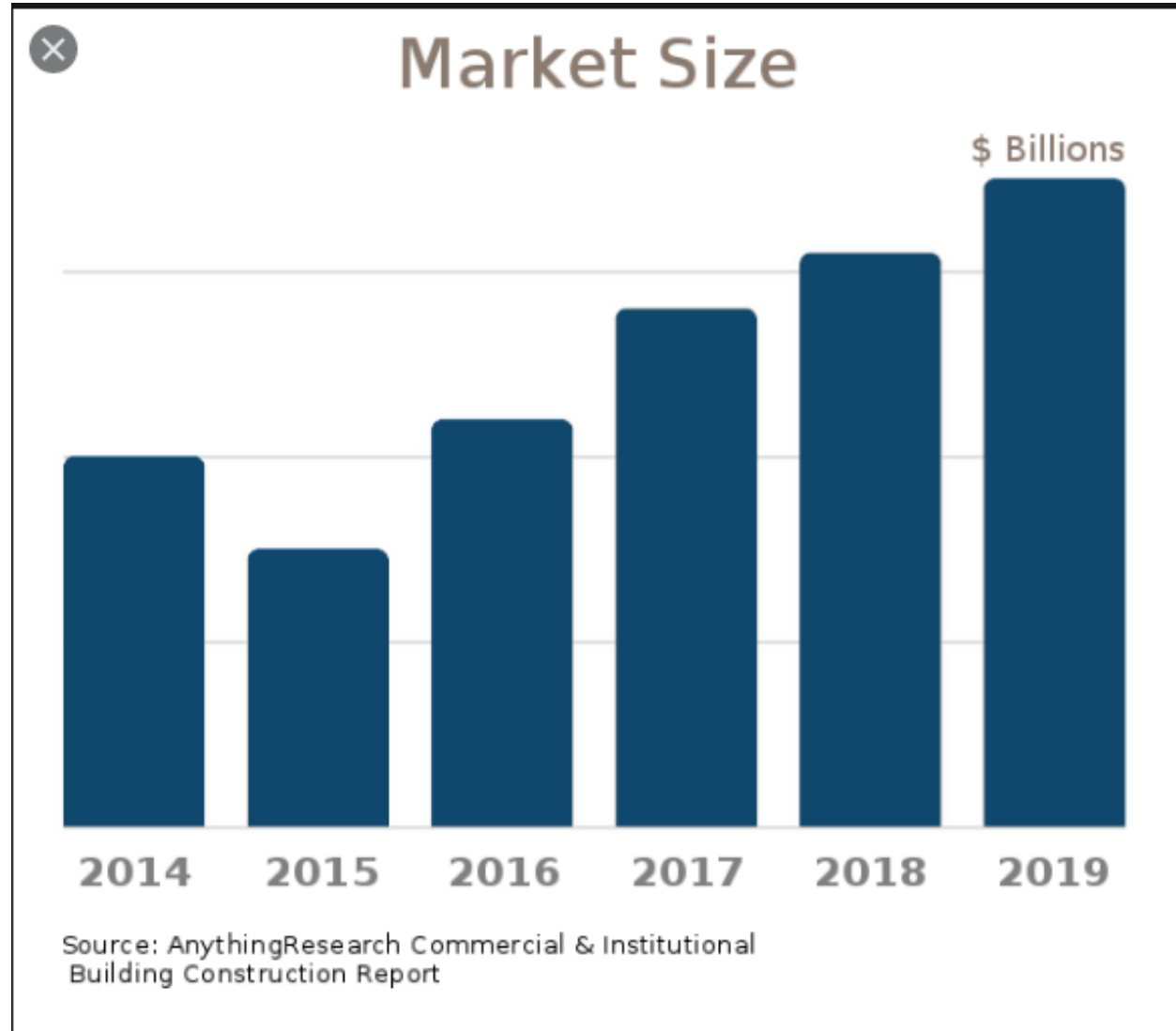
- The global construction market reached a value of nearly \$12,744.4 billion in 2019, having increased at a compound annual growth rate (CAGR) of 6.1% since 2015
- The market is expected to decline from \$12,744.4 billion in 2019 to \$12,633.9 billion in 2020 at a growth rate of -0.9%.

# Construction Market Size





# Construction Market Size



# Major Share of Construction Market



# Sector Wise Construction Market

| Sector/Type of Demand                    | ₹ Billion          |
|------------------------------------------|--------------------|
| <b>Public Sector</b>                     | <b>4.7 – 5.5</b>   |
| <i>Building Construction Sub-total</i>   | 3.0 – 3.7          |
| <i>Residential</i>                       | 1.1 – 1.3          |
| <i>Commercial</i>                        | 0.1                |
| <i>Industrial</i>                        | 0.1                |
| <i>Institutional &amp; Others</i>        | 1.7 – 2.2          |
| <i>Civil Engineering Works Sub-total</i> | 1.7                |
| <b>Private Sector</b>                    | <b>7.3 – 8.0</b>   |
| <i>Building Construction Sub-total</i>   | 6.8 – 7.5          |
| <i>Residential</i>                       | 2.6 – 2.7          |
| <i>Commercial</i>                        | 1.4 – 1.7          |
| <i>Industrial</i>                        | 2.3 – 2.6          |
| <i>Institutional &amp; Others</i>        | 0.5                |
| <i>Civil Engineering Works Sub-total</i> | 0.5                |
| <b>TOTAL CONSTRUCTION DEMAND</b>         | <b>12.0 – 13.5</b> |

# 1.3 Construction Project Characteristics

## Definition:

- Construction Implies **designing, building, installation and commissioning** of items of Civil, Mechanical, Electrical, telecommunication and other utility work

## Characteristics:

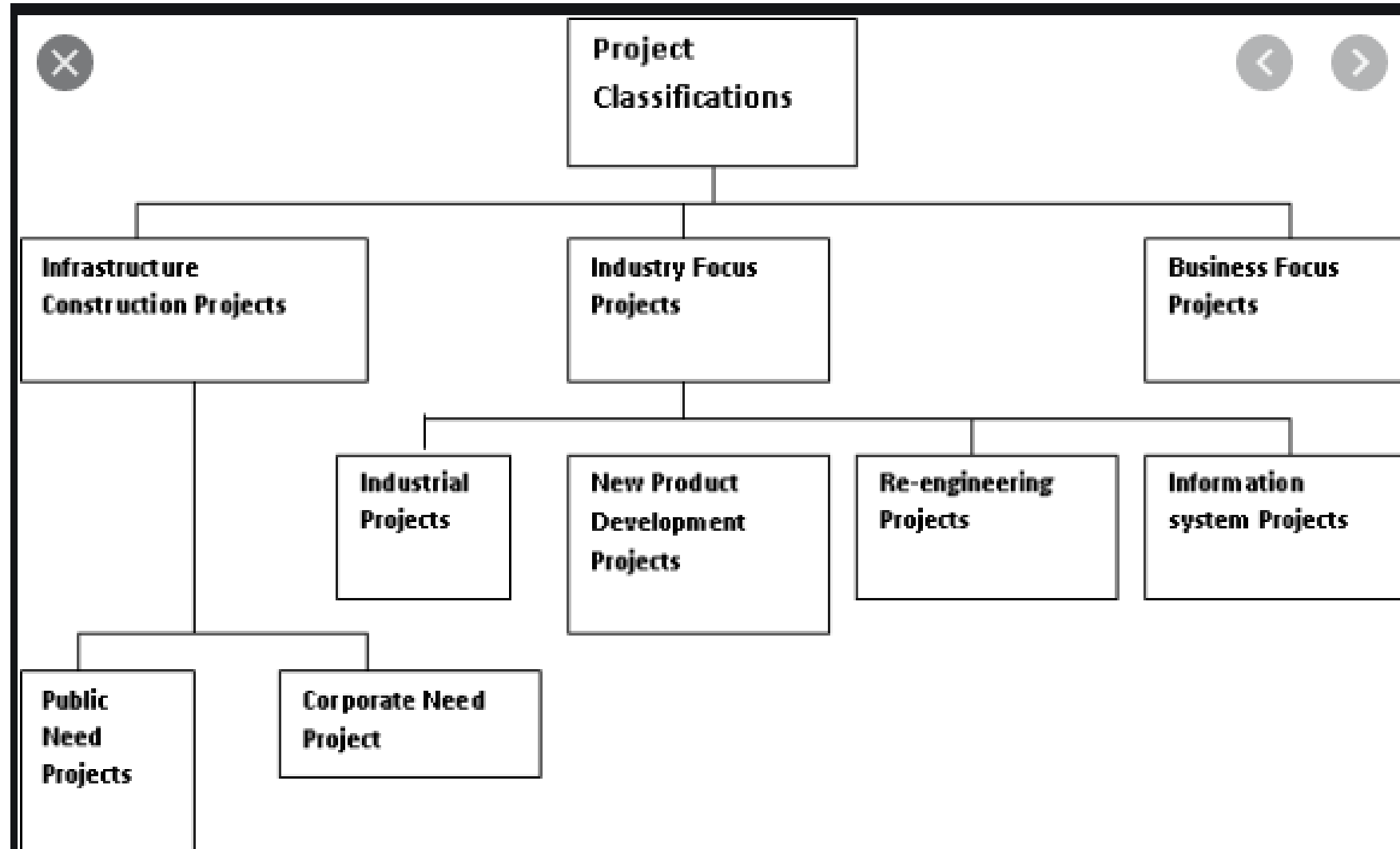
- High valued
- Time bounded
- **Predetermined objectives** in terms of quality, completion time, budgeted cost and other specified constraints
- Unique construction mission
- Complex project environments
- Putting together resources
- Temporary organization

## Construction project Characteristics:

- Headed by a project manager
- Numerous interdependent and interrelated activities
- Employ huge resources
- Heavy investment
- High degree of risk
- High level of technology management

### **Classification:**

# Construction Project Classification



# Construction Project Classification

## **1) Nature of construction Facility**

- Building
- Infrastructure
- Industrial
- Specific Purpose

## **2) Nature of work**

- Repetitive
- Non-repetitive work

## **3) Mode of execution**

- Departmental
- Contractual

# Construction Project Classification

## 4) **Nature of construction contract**

- Cost plus
- Item rate
- Lump sum
- Turnkey or BOT (built operation and Transfer)

## 5) **Completion time**

- Long duration (over 5 yrs)
- Medium duration (3-5 yrs)
- Normal duration (1-3 yrs)
- Short duration (less than 1 yr)



# Construction Project Classification

6) **Budgeted cost** (as per Indian public sector central Government project classification)

- Mega project (over 1000 cores)
- Large project (100 - 1000 cores)
- Medium project (20 - 100 cores)
- Small project (less than 20 cores)

7) **Maturity level**

- Insignificant risk
- Low value risk
- Medium value risk
- High value risk

# Construction Project Classification

## **8) Need based project**

- Public
- Corporate
- Commercial
- Re-engineering

## **Construction Project Performance Objectives:**

- Scope
- Quality
- Resources
- Completion time
- Cost

# Construction Project Performance Objectives

## **Main parameters:**

- Scope
- Quality
- Resources
- Completion time
- Cost

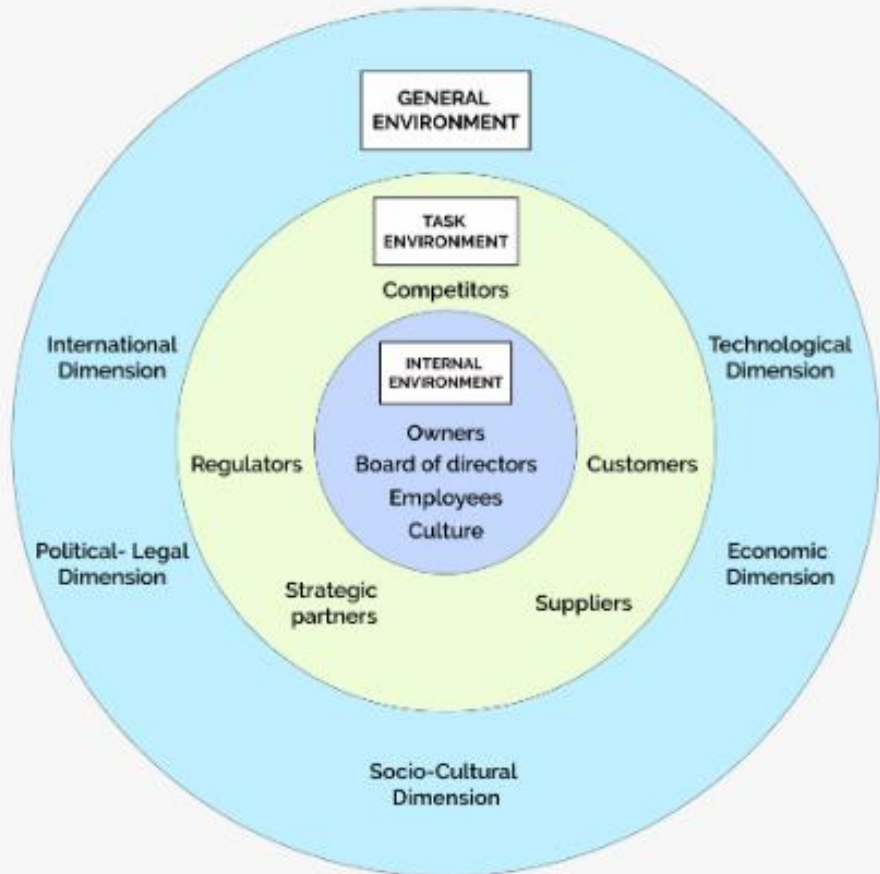
## **Other parameters:**

- Organization, Culture, Stakeholder's interest, Contractor's performance capability

**Each parameter is a function of others**

# Construction Project Environment

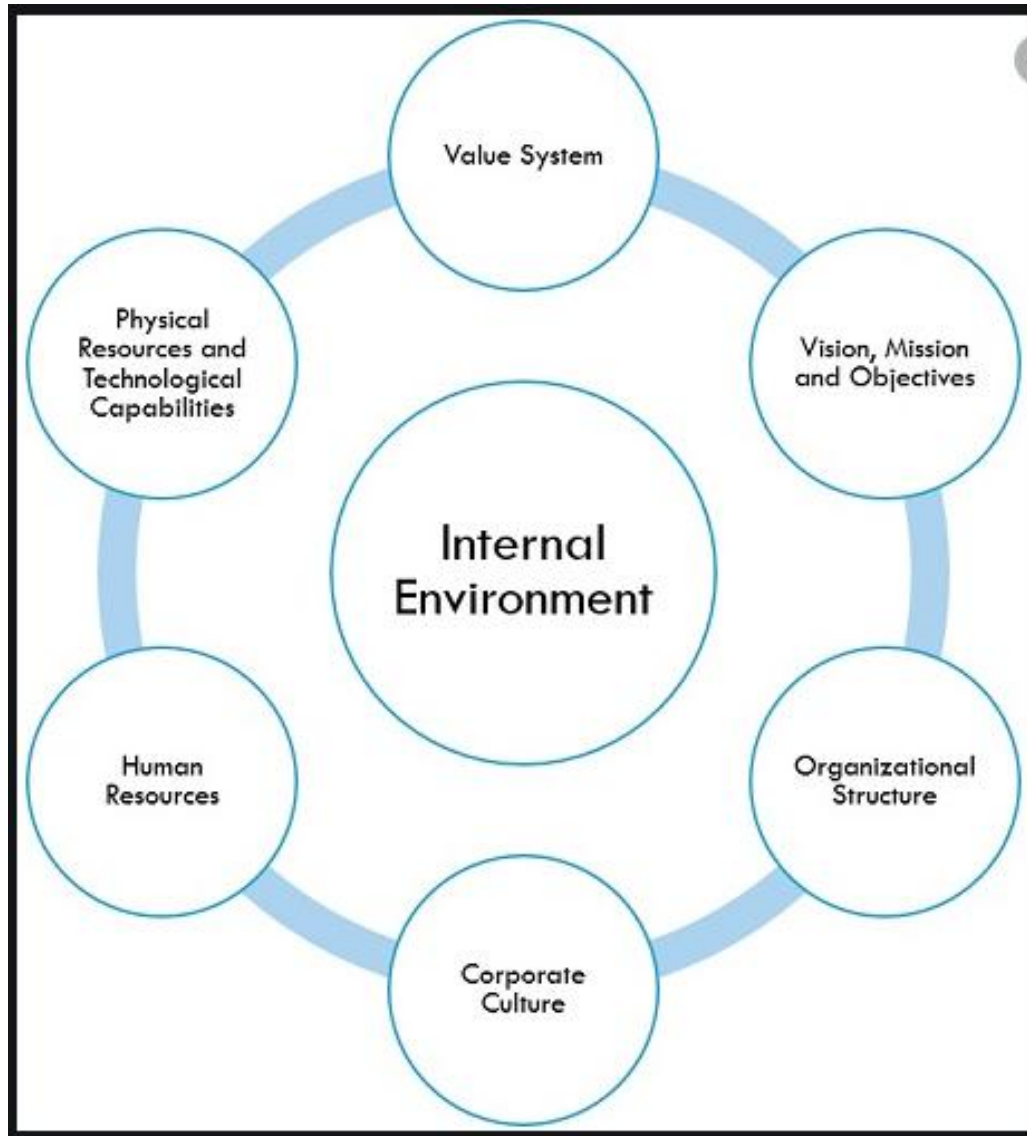
## Elements of Organizational Environment



## Task Environment

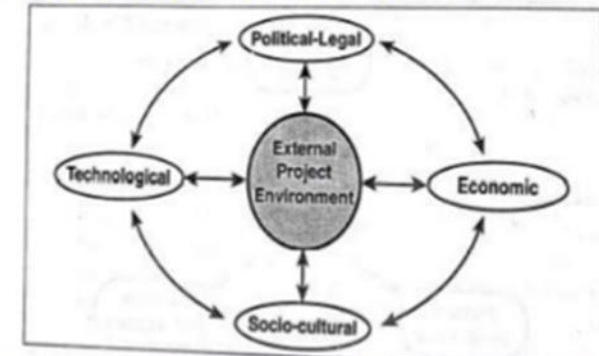
- Immediately surrounds Project
  - Made up of Stakeholders
  - Interest and Impacts are Interrelated
  - Affect Project Activities
  - Project can Influence Task Environment
- They are:
- Client
  - Contractor
  - Consultant
  - Competitor
  - Suppliers
  - Government
  - Labor
  - Unions
  - Financers
  - Owner
  - Shareholders
  - Organization
  - Donor

# Construction Project Environment

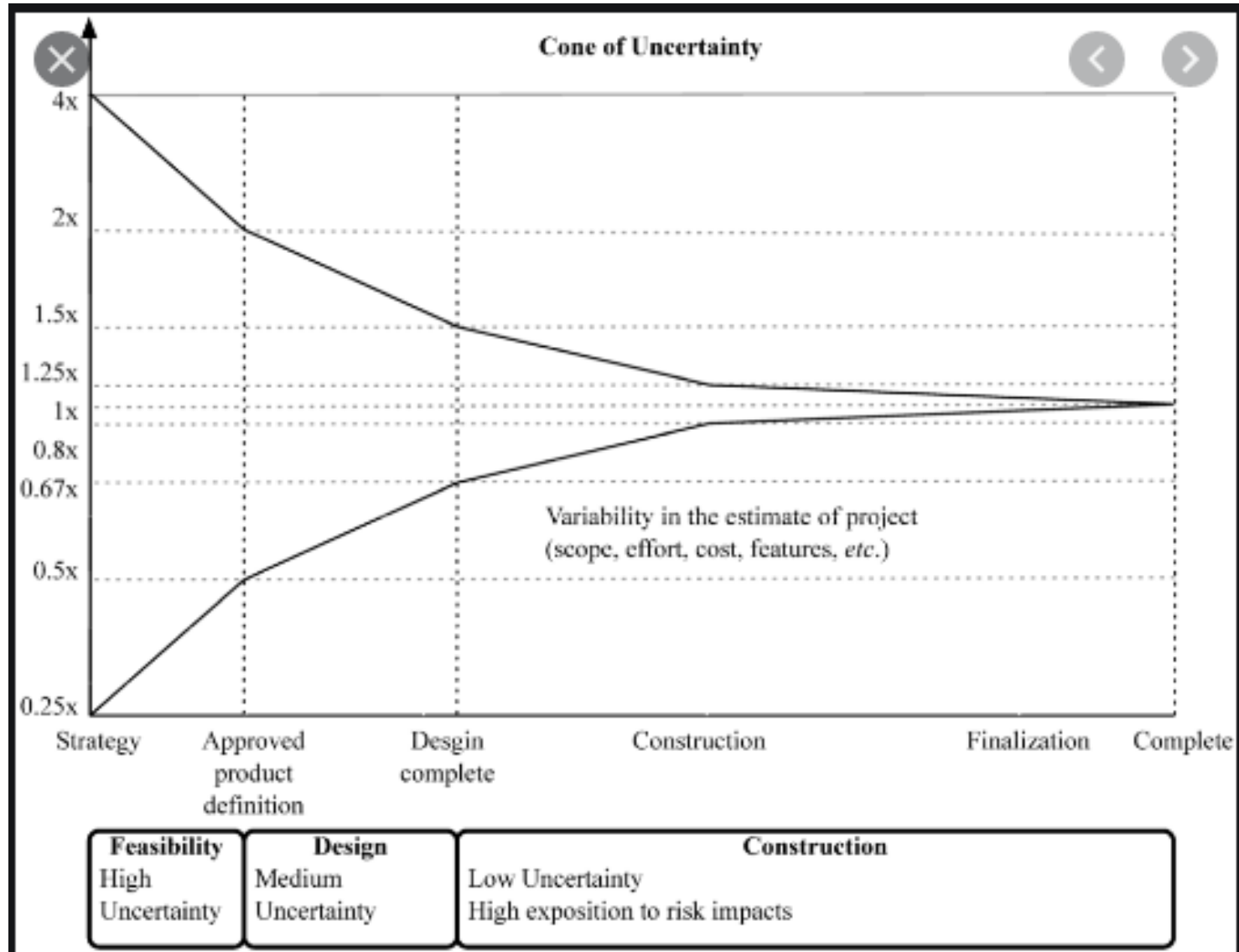


## External Environment:

- It is located outside the project.
- It cannot be controlled by the project.
- It provides opportunities and threats to the project.
- The forces in the external environment are:
  - ✓ Economic
  - ✓ Technological
  - ✓ Political-Legal
  - ✓ Socio-cultural



# Uncertainty in Construction Project



# Uncertainty in Construction Project

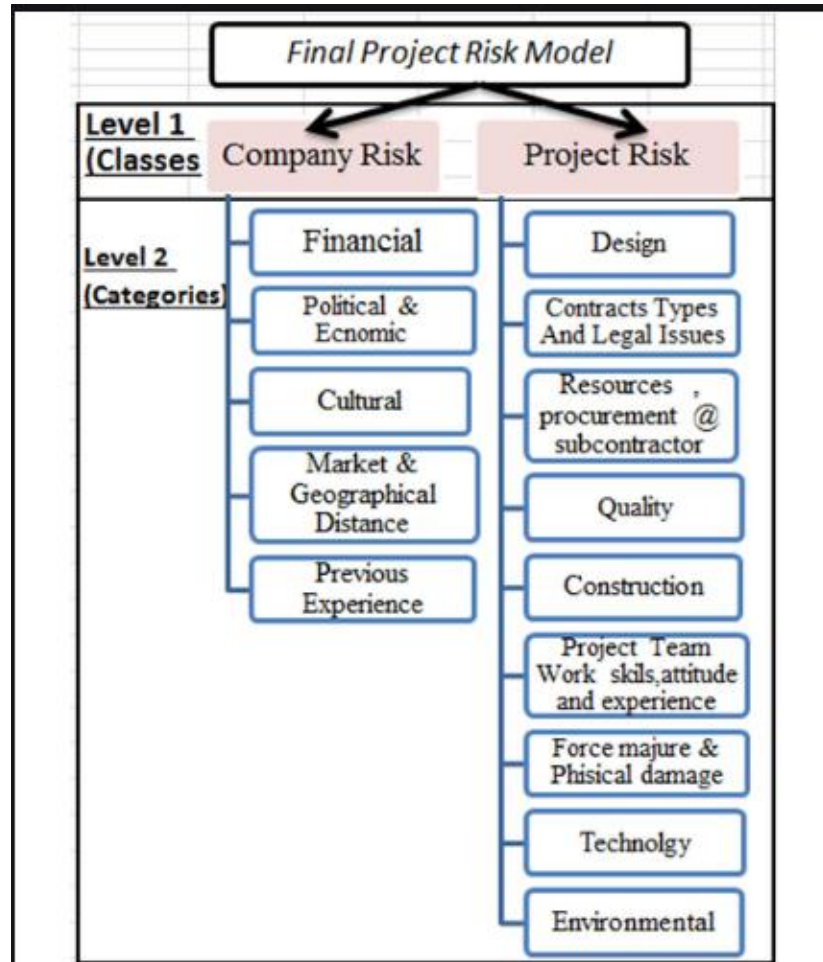
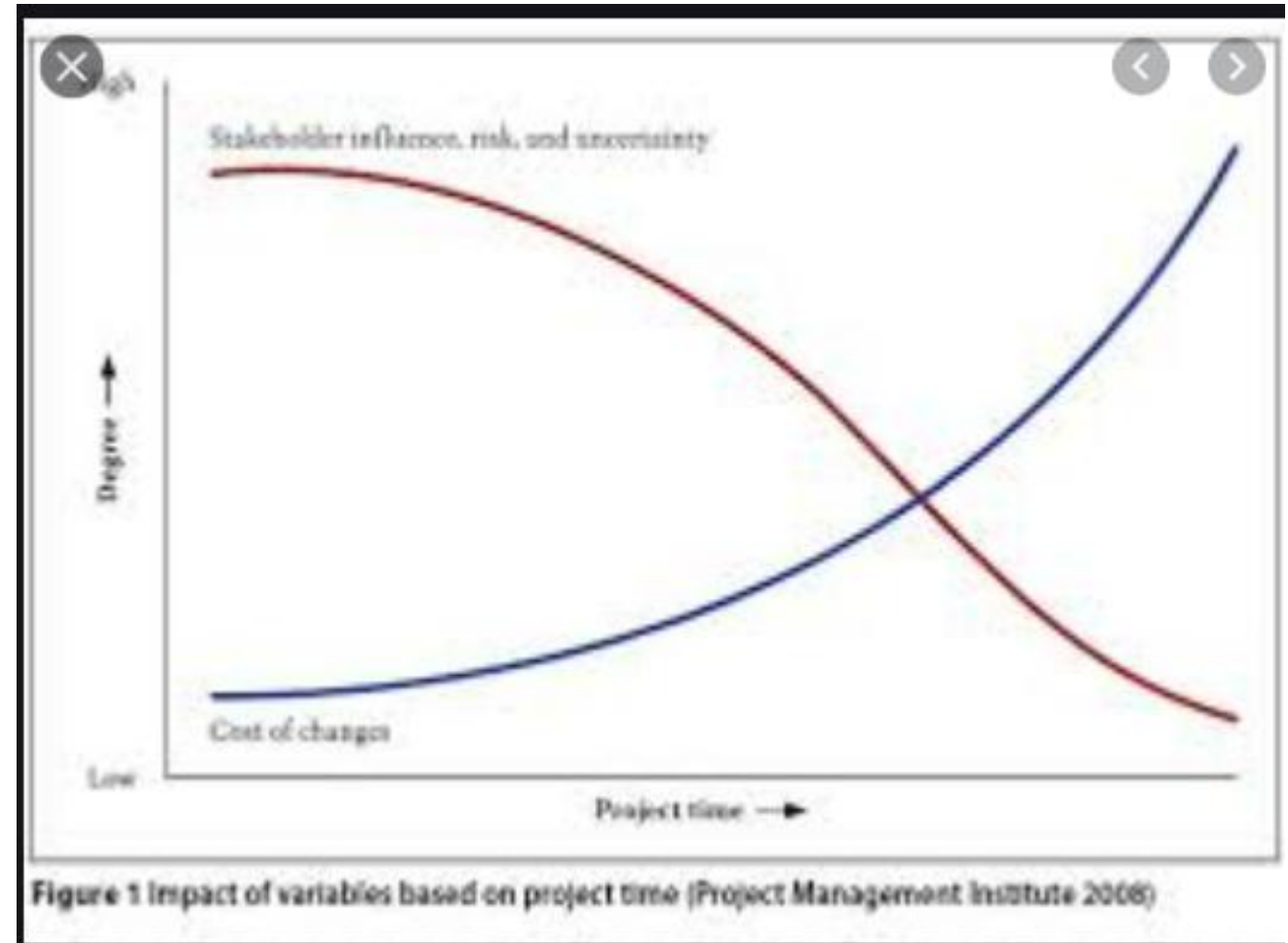
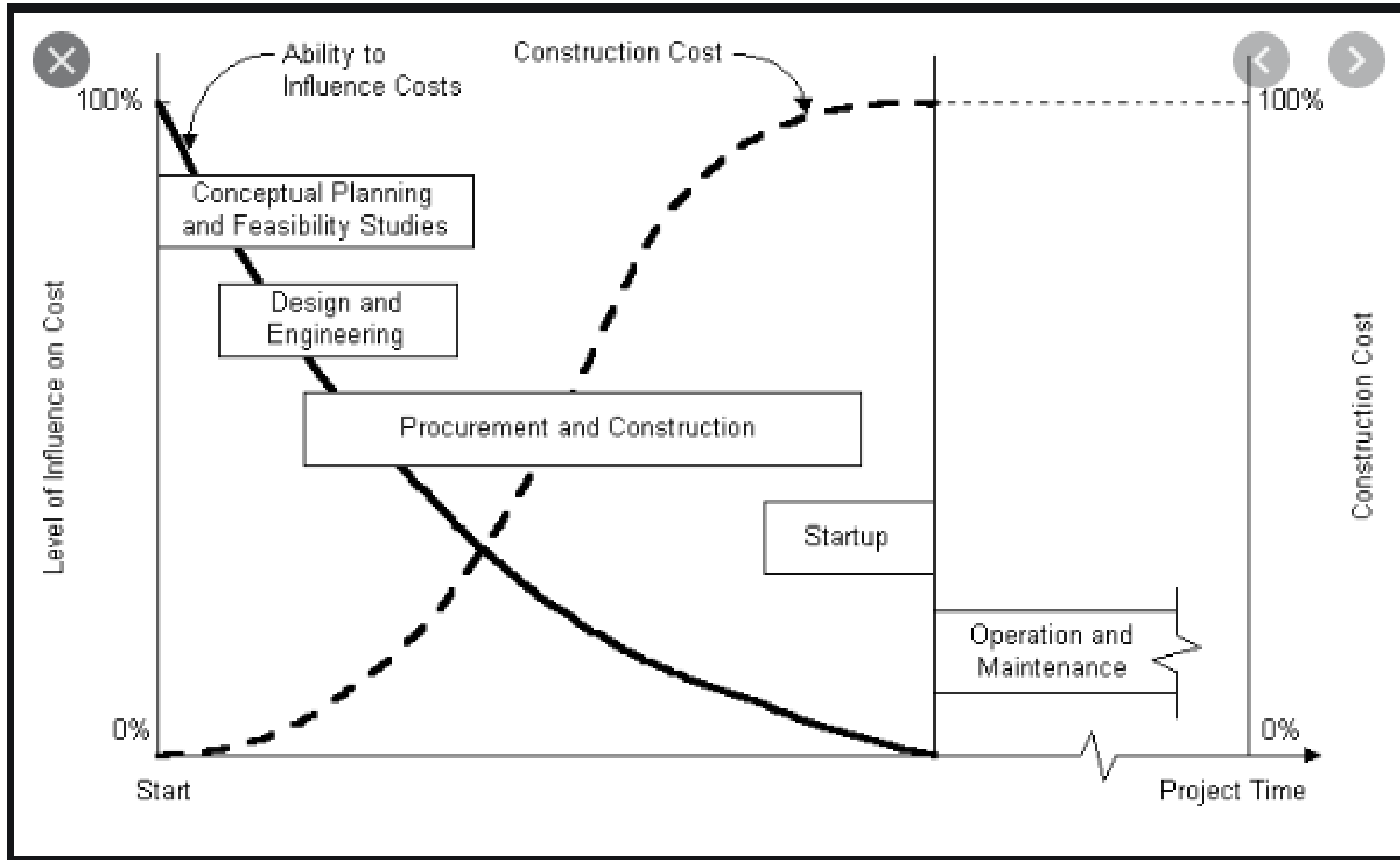


Fig. 2. Risk hierarchy model in company and project Levels

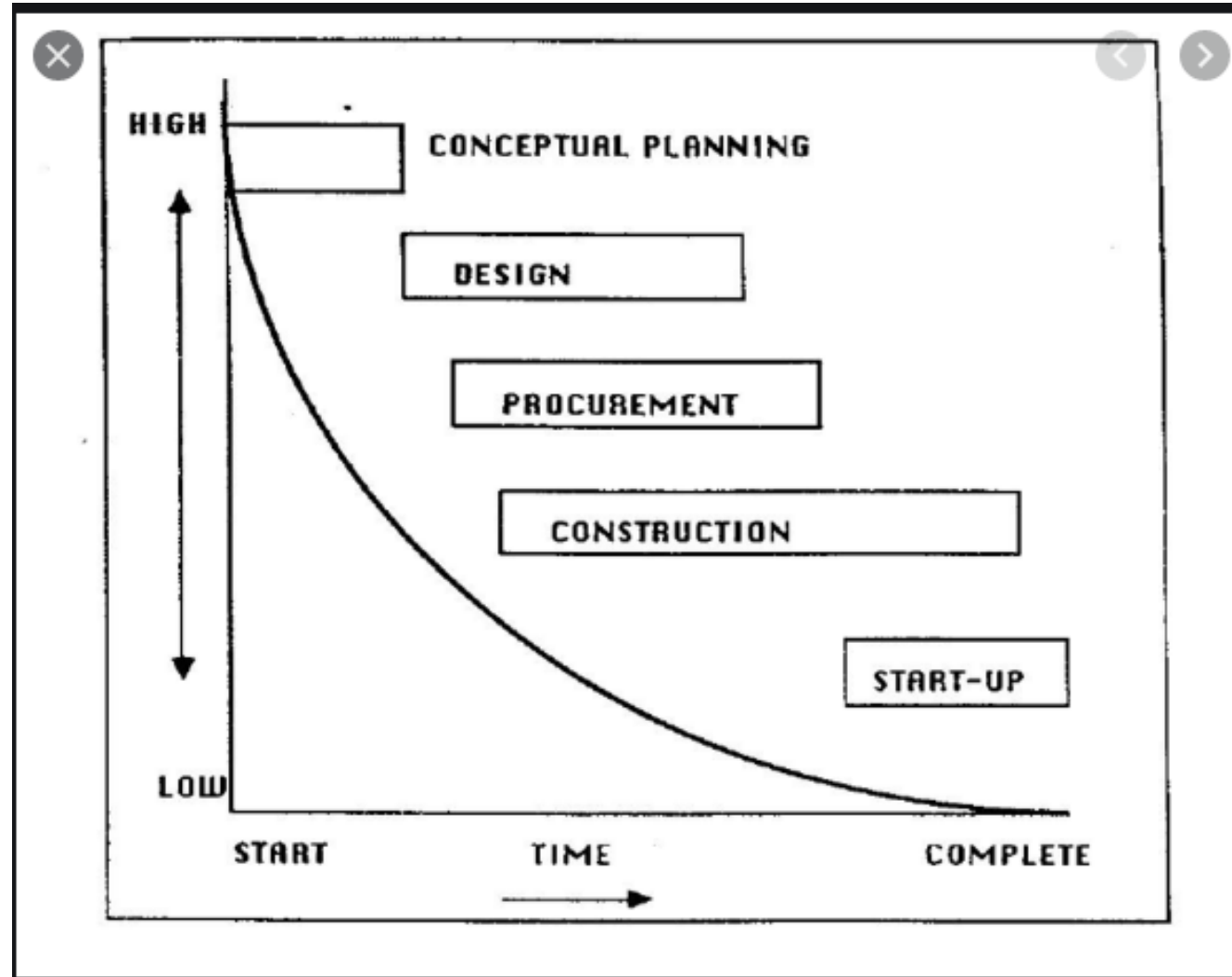


# Construction Project Common Attributes

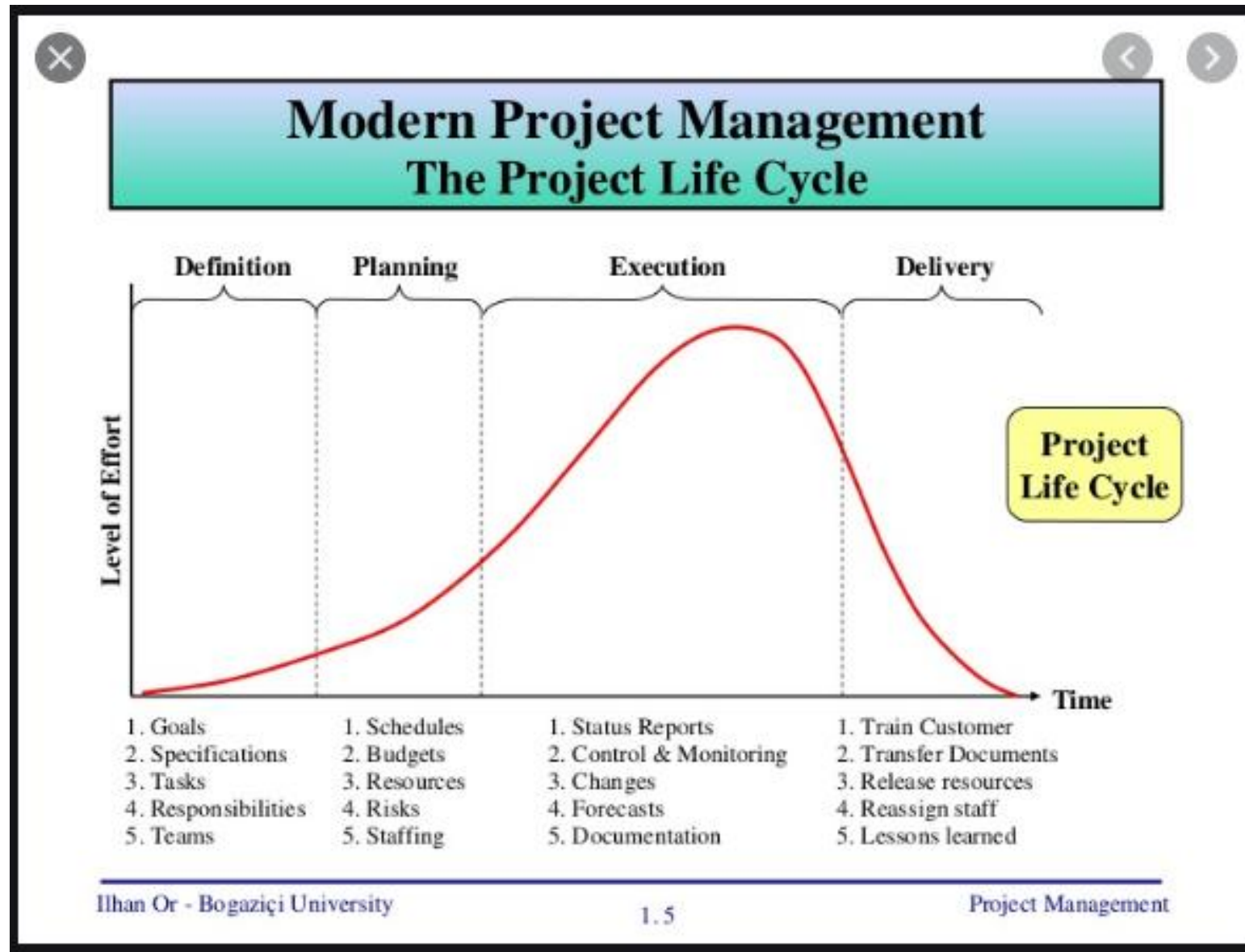




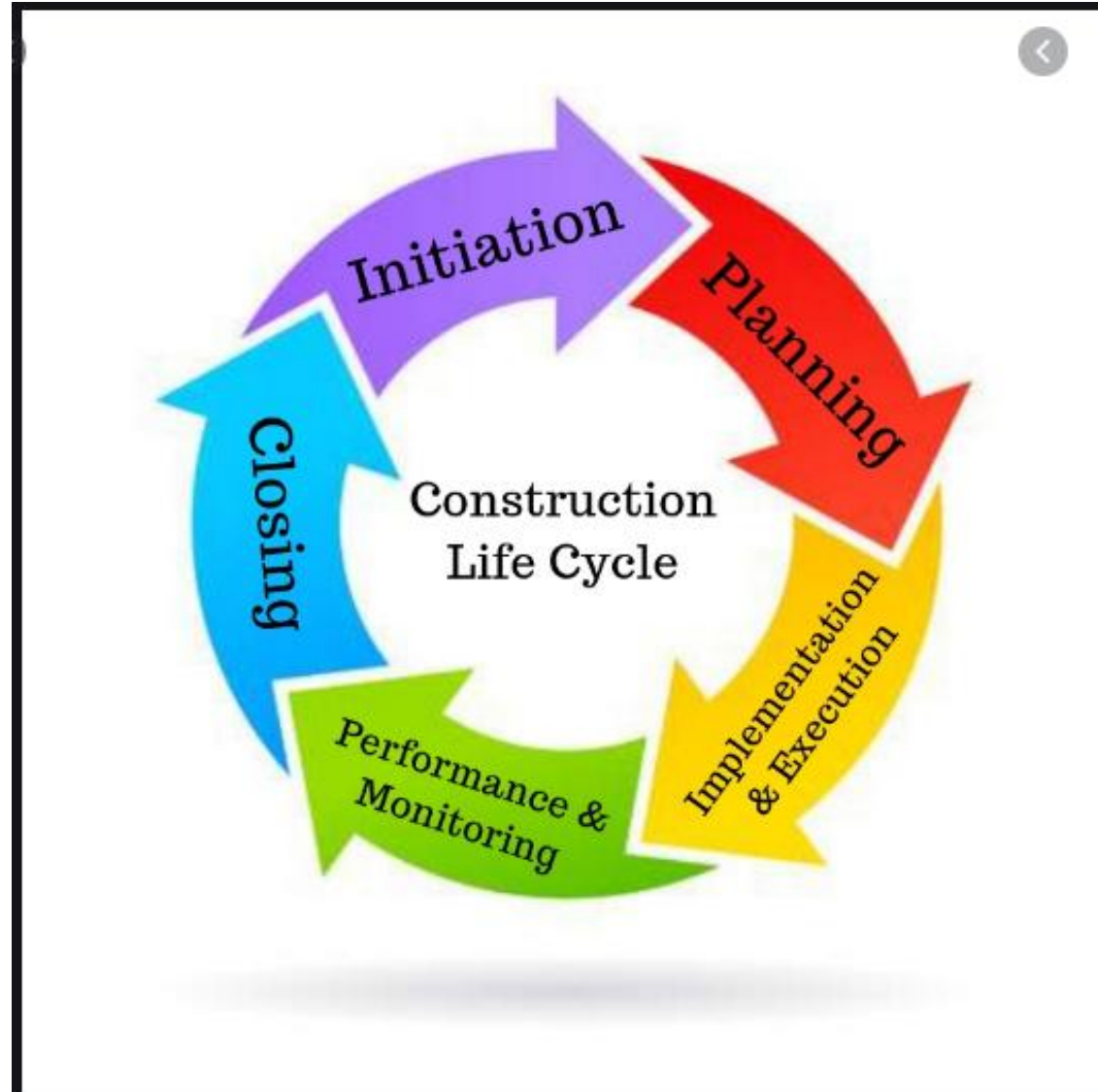
# Ability to Influence Final Cost Over Project life



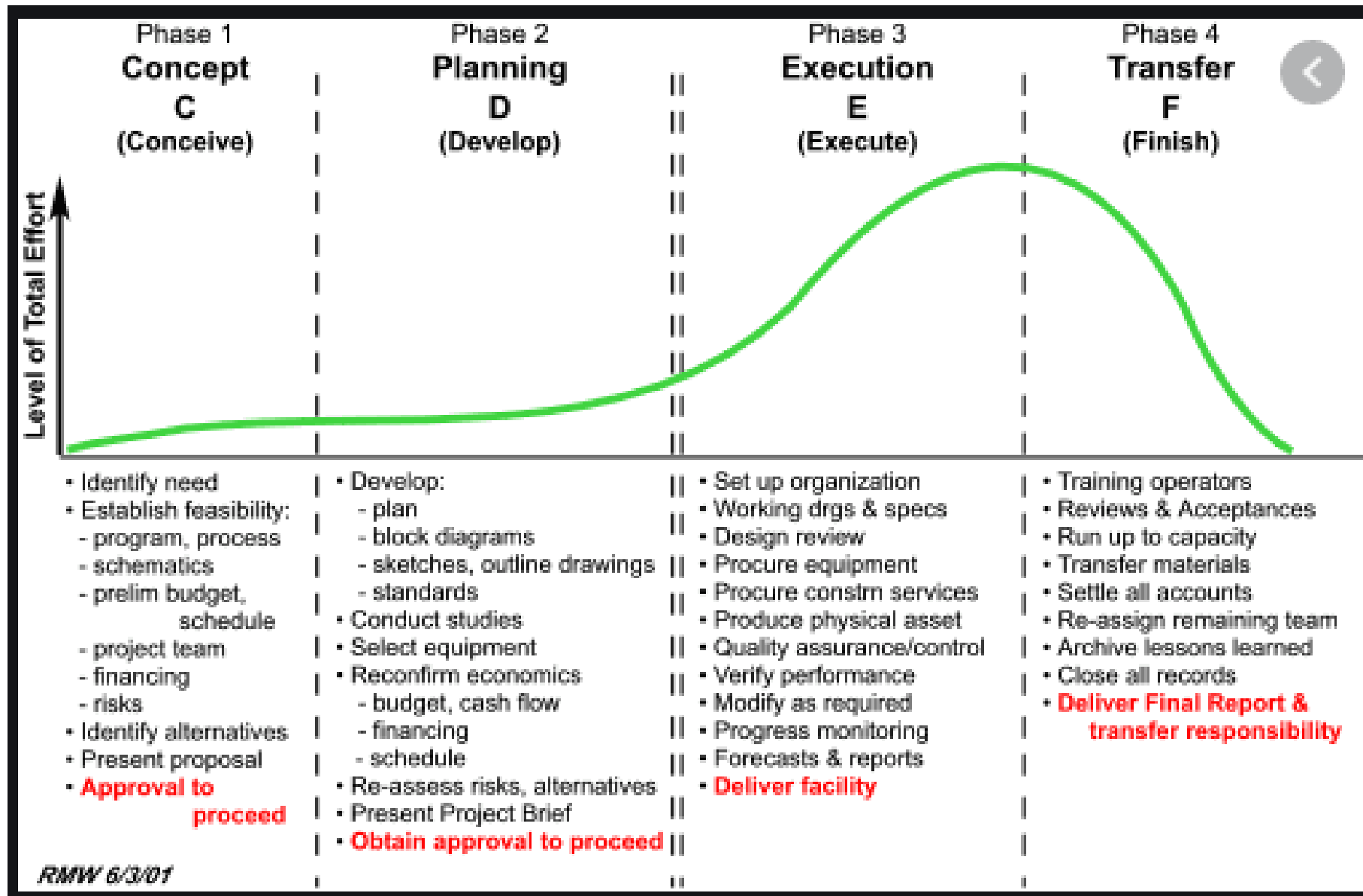
# Project Life Cycle



# Construction Life Cycle



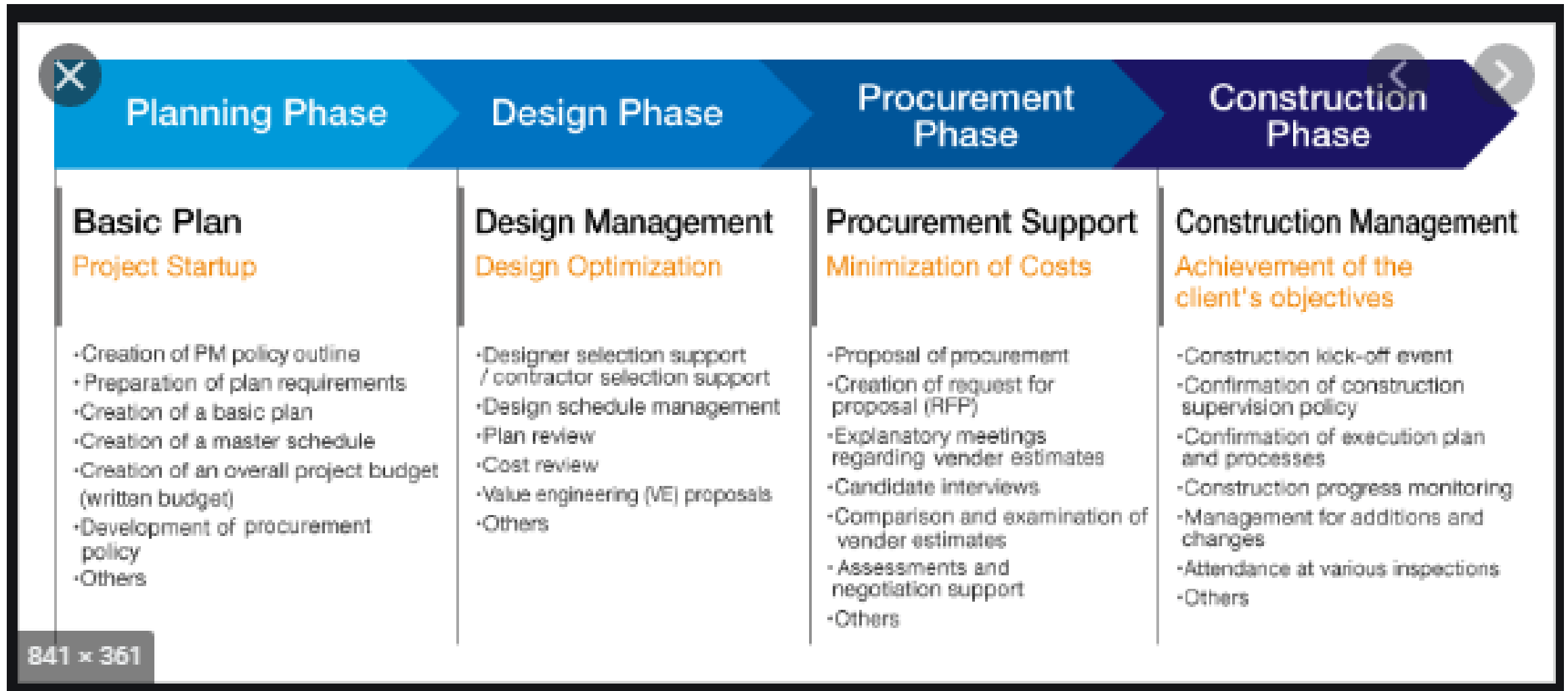
# Construction Project Phases



# Construction Project Management Phases



# Construction Project Management Phase



# Phase Wise Responsibility in Construction Project

|  No. | Phase                     | Entities Responsible   |
|---------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                                     | Feasibility               | Owner                                                                                                                                                                                        |
| 2                                                                                     | Design                    | Owner, design office                                                                                                                                                                         |
| 3                                                                                     | Construction              | Owner, design office, contractor                                                                                                                                                             |
| 4                                                                                     | Operation and Maintenance | Owner, maintenance contractor                                                                                                                                                                |
| 5                                                                                     | Decommissioning           | Owner                                                                                                                                                                                        |

611 x 218

# Project Management in a Rational Unified Process

| 1.0 Project Initiation  | 2.0 Project Solution Planning | 3.0 Solution Implementation           | 4.0 Project Closure  |
|----------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| Develop Opportunity Profile                                                                              | Structure Technical Solution  | Manage Contract and Vendor Agreements | Close Out Contract and Vendor Agreements                                                                |
| Define Project                                                                                           | Select Vendors                | Manage Resources                      | Close Out Project Documents                                                                             |
| Develop Business Case                                                                                    | Prepare Project Plan          | Manage Scope Changes                  | Disperse/Reassign Project Resources                                                                     |
| Initiate Project                                                                                         | Structure Business Solution   | Track and Control Progress            |                                                                                                         |
|                                                                                                          | Establish Customer Contract   | Manage Quality and Acceptance         |                                                                                                         |
|                                                                                                          | Prepare for Project Execution | Pre-Close Out Review                  |                                                                                                         |
| 675 x 358                                                                                                |                               |                                       |                                                                                                         |



# Planning Process



# Planning Phase of Construction Project

## Planning Phase

### Key Outputs of this phase are:

- ❑ Create WBS and WBS Dictionary
- ❑ Develop Schedule
- ❑ Develop Budget
- ❑ Determine roles and Responsibilities
- ❑ Develop Communication Plan
- ❑ Develop Risk Plan
- ❑ Create Performance measurement baselines
- ❑ Develop Preliminary Scope Statement
- ❑ Test Plan
- ❑ Determine the Critical Path
- ❑ Prepare procurement documents
- ❑ Create Process Improvement Plan
- ❑ Gain Formal Approval of all plans and key deliverables

\* After that you as the project manager are good to Launch Kickoff Meeting!

# Closing Process

 **Project Management Processes**

 Master of Project Academy

## Closing Process Group Activities

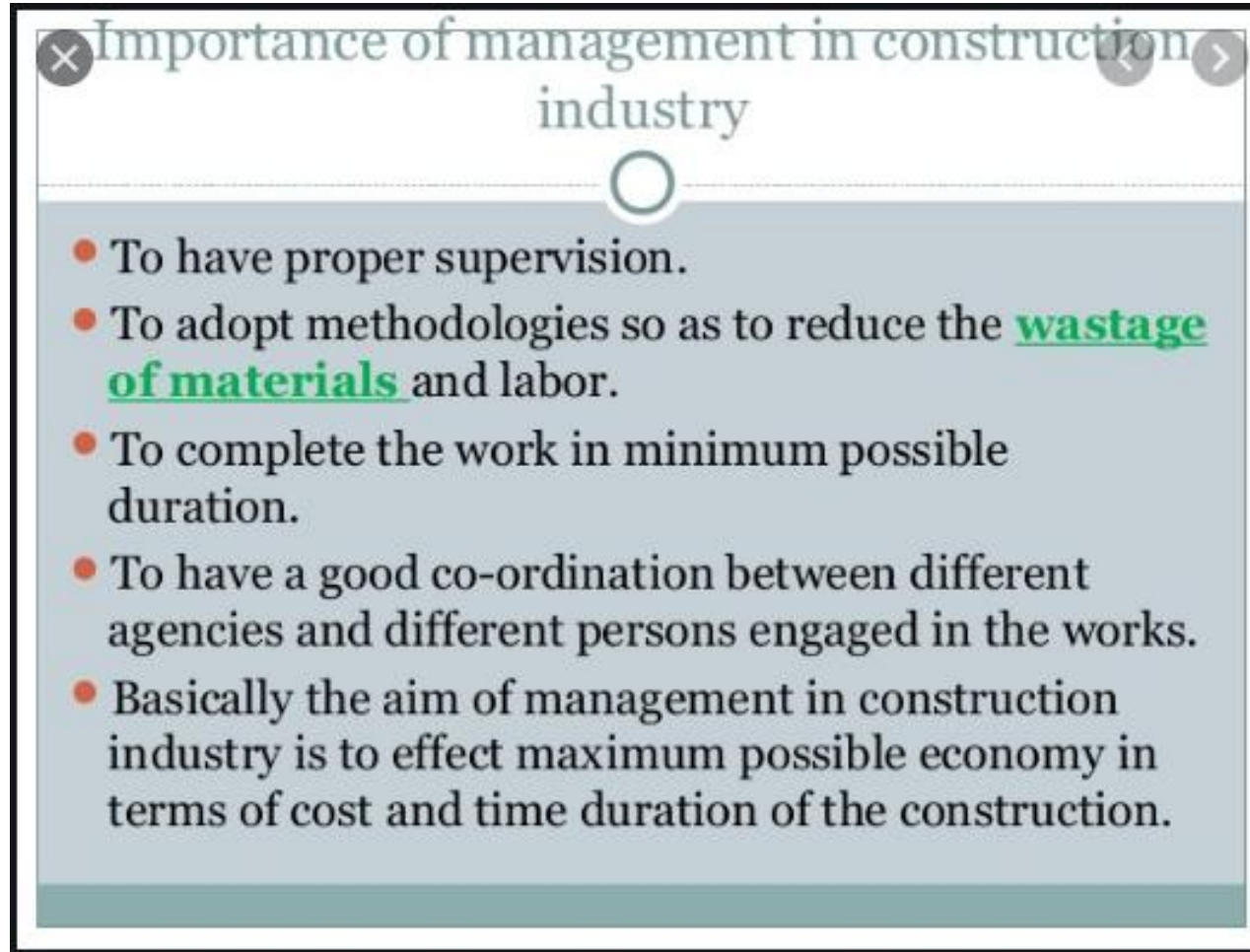
- Confirm work is done as per requirements
- Complete procurement closure
- Gain formal acceptance
- Complete final performance reporting
- Index and archive records
- Update lessons learned



1024 x 576

# 1.5 Construction Project Management

## Necessity of Construction Management



Importance of management in construction industry

- To have proper supervision.
- To adopt methodologies so as to reduce the wastage of materials and labor.
- To complete the work in minimum possible duration.
- To have a good co-ordination between different agencies and different persons engaged in the works.
- Basically the aim of management in construction industry is to effect maximum possible economy in terms of cost and time duration of the construction.

# Functions of Management

**Planning:** To answer the following question is planning.

- What to do?
- When to do?
- How to do?

**Organizing:** It is the process of establishing a **structural relationship** among the functions of people so as to streamline the objectives.

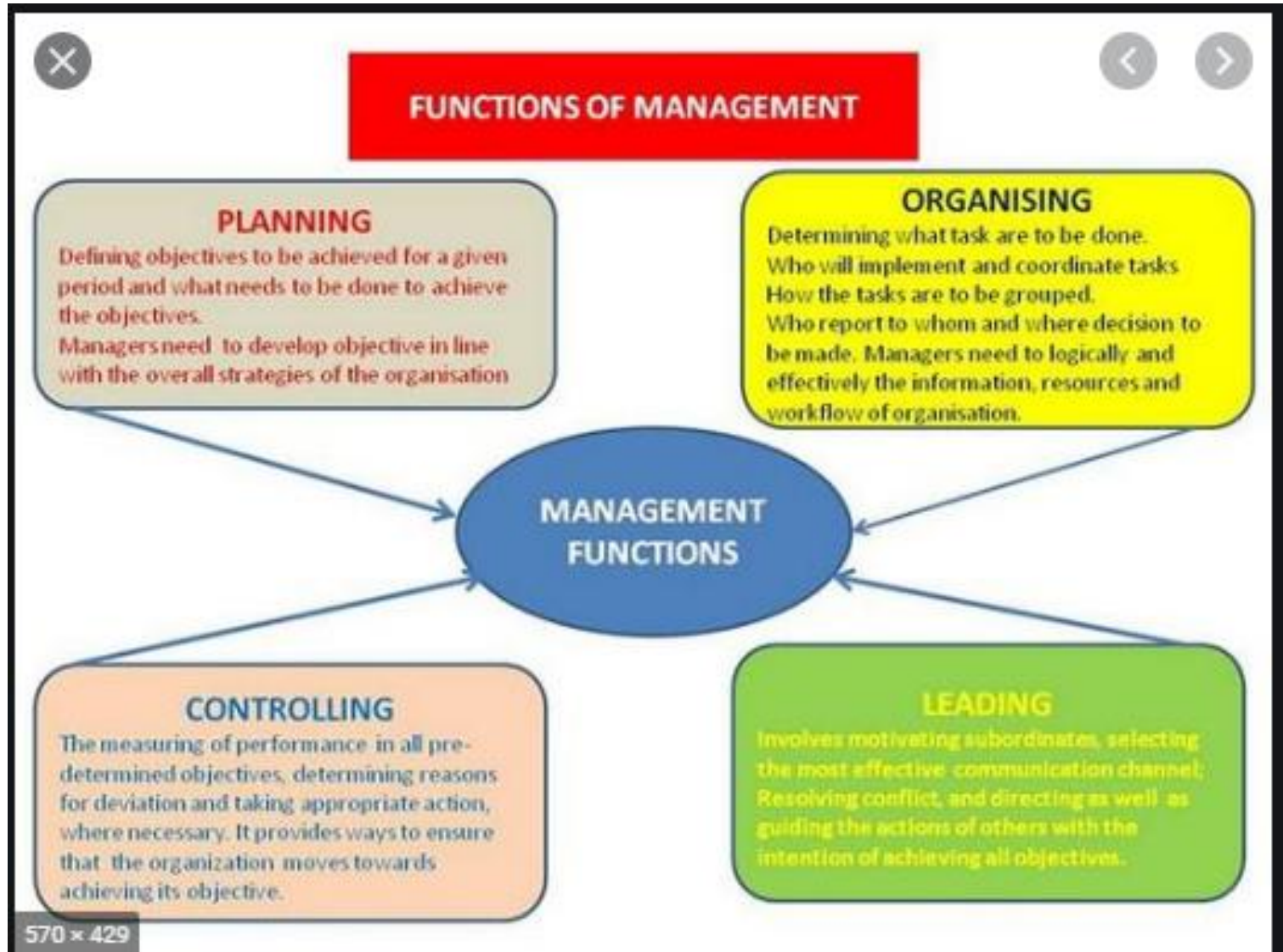
**Staffing:** **Selection of right person** to different job.

**Directing:** It involves **motivation, guidance, supervision and leading** the employees of the organization.

**Controlling:** It is the **monitoring of progress** achieved in comparison with planned program and to identify the areas of deficiency so that remedial measures may be taken.



# Functions of Management



# Scientific Method of Construction Management

## **Aim of scientific management:**

- To ensure maximum productivity
- To reduce the risk

It applies various tools and techniques (CPM, PERT, Time-cost-trade off etc)

## **Principles of Scientific Management:**

- Work study
- Task planning
- Selection and training of staffs
- Tools and materials
- Management and workers relation

## 1.6 Relationship between Client, Consultant and Contractor

- There must be **team work approach** among the three parties
- Each party must have supportive role with another
- Each of them must follow their role and responsibility
- Owner must provide the sufficient fund on time
- Consultant must prepare the design as per interest of Owner
- Contractor should complete the execution work maintaining quality and schedule saving cost